Lecture 6: Assumptions, standard errors and changing samples

Empirical Methods for Law and Economics | EDLE

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Today's questions

When can we trust a standard error?

What changes when one country, or the sample, changes?

- By the end of today you will be able to:
  - state the assumptions behind a regression and what fails when each one fails;
  - detect unequal error variance and report robust standard errors;
  - find observations that drive a result and decide what to do with them;
  - compare models on the same sample;
  - write a referee comment on a published table.

DISCUSS Two claims in one sentence

"With income held fixed, -2.04 is the effect of police on theft, and $p = 0.057$ tells us how sure we can be."

Questions:

1. This sentence makes two different claims. Which ones?
2. For each claim, what must be true about the residual u_i (everything that affects theft and is not in the model)?

ANSWERS Two claims in one sentence

- Claim 1, *the coefficient is the effect*: nothing left in u_i may be related to police. Yesterday we saw why this fails (reverse causality, other confounders, recording).

- Claim 2, *the p-value measures our uncertainty*: the standard error must be computed correctly. That requires assumptions on u_i too: independent observations and, for the usual formula, the same variance for all countries.

- The two claims fail for different reasons. Today: mostly claim 2, and what one country or a change of sample can do to both.

The assumptions behind a regression

Assumption	In words	If it fails
1. Linear in the coefficients	y is a straight-line function of the x's (logs allowed)	the line summarises the data badly
2. Independent observations	one country tells us nothing about the error of another	standard errors usually too small
3. No perfect collinearity	no x is an exact combination of the others	Stata drops a variable
4. Zero conditional mean	nothing in u_i is related to the x's	coefficients biased: no causal reading
5. Constant variance	u_i has the same spread for every value of x	usual standard errors wrong; coefficients fine
6. Normal errors	u_i follows a normal distribution	matters only for exact t in small samples

- Assumption 4 is about the design, not the software. Assumptions 2 and 5 are about the standard errors.

DISCUSS Which assumption fails?

For each situation, name the assumption (1–6) that is in doubt.

1. More police officers make victims more likely to report a theft, so recorded theft rises with police.
2. Ten day-care centres observed for 20 weeks: the analysis treats the 200 centre-weeks as independent.
3. Recorded theft varies much more among rich countries than among poor ones.
4. A regression includes both `high_income` and `not_high_income` (equal to 1−`high_income`) together with a constant.
5. Enforcement time rises with income for poor economies and falls for rich ones.

ANSWERS Which assumption fails?

- (1) Assumption 4: the recording error is related to police. The coefficient is biased upward.
- (2) Assumption 2: weeks of the same centre share parents and staff. Ten independent units, not 200.
- (3) Assumption 5: unequal variance. The coefficient is fine; its usual standard error is not.
- (4) Assumption 3: perfect collinearity. Stata drops one of the variables.
- (5) Assumption 1: the relation is not a straight line. A single slope averages two opposite patterns.

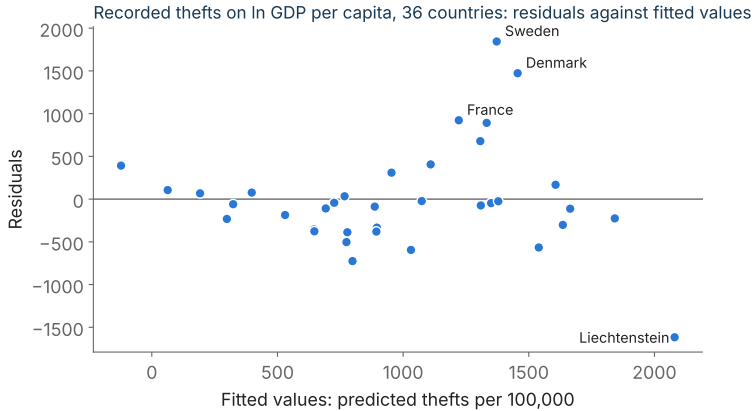
PREDICT Does the spread stay the same?

- Regression: recorded thefts per 100,000 on ln(GDP p.c.), 36 countries.
- After the regression, each country has a fitted value and a residual.

Predict: will the residuals be equally spread for countries with low and high fitted values? Why?

- Think about what makes recorded theft high: reporting, insurance, police recording practices.

Residuals against fitted values



- The spread grows with the fitted value: residuals are small where predicted theft is low, large where it is high. This is **heteroskedasticity**.

STATA Guided Stata 1: unequal variance and robust standard errors

```
use "$course_data/justice_europe_2019.dta", clear
generate ln_gdppc = ln(gdp_pc_usd)
regress theft_100k ln_gdppc
rvfplot, yline(0)
estat hettest
regress theft_100k ln_gdppc, vce(robust)
```

Slope of ln GDP p.c. ($n = 36$)	Coef.	SE	t	95% interval
Usual standard errors	606.4	123.5	4.91	355 to 857
Robust standard errors	606.4	146.9	4.13	308 to 905

- `estat hettest` (Breusch–Pagan): $\chi^2(1) = 12.71$, $p < 0.001$: constant variance is rejected.

Robust standard errors

- Same coefficients; different standard errors, t, p-values and intervals.
- Robust (heteroskedasticity-robust, "White") standard errors do not require constant variance. If the variance is constant, they give similar answers.
- They are the default in applied work with cross-section data. No test is needed to use them: a test that fails to reject, like any test, does not prove the assumption.
- What they do *not* fix:
  - dependence between observations (ten centres, 200 rows): that needs standard errors that count independent units;
  - a biased coefficient (assumption 4).

DISCUSS Footnote 10, again

- Voigt and Wulf (footnote 10): "The result of the Breusch–Pagan and Cook–Weisberg test was not significant (p-value 0.451). The result shows that our model does not suffer from heteroscedasticity."

Questions:

1. The test we just ran on recorded theft rejected constant variance with 36 countries. Theirs did not, with 74. What can you conclude about their model?
2. What would you ask the authors to do instead? Is it costly?

Voigt, S. and Wulf, A.J. (2019), *Journal of Institutional Economics* 15(1): 99–120, footnote 10, p. 112.

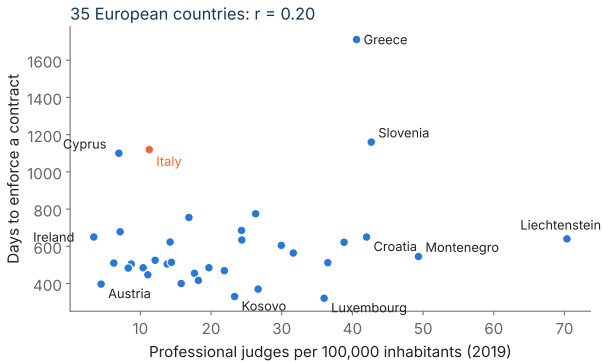
ANSWERS Footnote 10, again

- Only that the data do not contradict constant variance. With 74 countries the test has limited power; "not significant" does not show that the problem is absent.

- Ask for robust standard errors in Table 2. It costs one option, `vce(robust)`, and it removes the need for the test.

- Referee comment: "Please report heteroskedasticity-robust standard errors in all columns of Table 2 and state whether any conclusion changes."

PREDICT One country



- Lecture 4: days to enforce a contract on judges per 100,000, 35 countries. Slope: 3.6 days per judge per 100,000.

Predict: which single country, if removed, would change the slope the most? In which direction?

Outliers, leverage and influence

- **Outlier**: a large residual; the country is far from the line (Greece: 1,711 days).
- **Leverage**: an unusual value of x; the country is far from the other countries horizontally (Liechtenstein: 70 judges per 100,000, the median is 18).
- **Influence**: removing the country changes the coefficient a lot. Influence needs both a large residual and some leverage.
- **Cook's distance** summarises influence for each observation. A common flag: Cook's distance above $4/n$.

STATA Guided Stata 2: influential observations

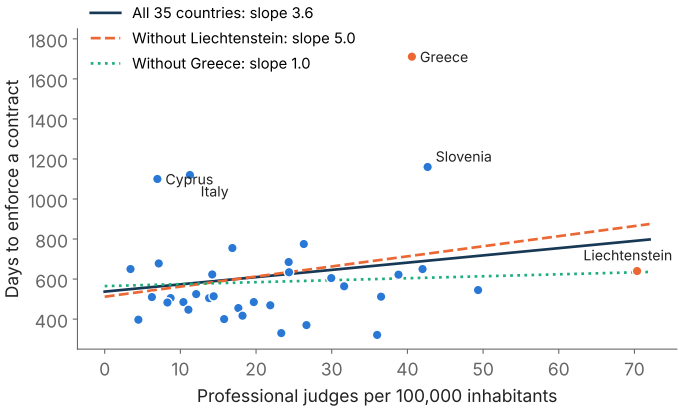
```
regress time_days judges_100k
predict lev, leverage
predict cook, cooksd
list country judges_100k time_days lev cook if cook > 0.1 & !missing(cook)
regress time_days judges_100k if country != "Greece"
regress time_days judges_100k if country != "Liechtenstein"
```

Country	Leverage	Cook's D
Greece	0.07	0.58
Slovenia	0.08	0.14
Cyprus	0.06	0.13
Liechtenstein	0.33	0.11

Sample	Slope	SE	p
All 35	3.62	3.16	0.26
Without Greece	0.98	2.42	0.69
Without Liechtenstein	5.04	3.83	0.20

- $4/n = 4/35 = 0.114$. Liechtenstein has by far the highest leverage, but Greece has the most influence.

One country can change the answer



- Without Greece the slope falls from 3.6 to 1.0; without Liechtenstein it rises to 5.0. None of the three is statistically different from zero.

DISCUSS Should we drop Greece? Liechtenstein?

Questions:

1. A colleague drops Greece "because it is an outlier" and reports a slope of 1.0. What is wrong with this?
2. Liechtenstein has 39,000 inhabitants. Is "judges per 100,000" comparable with the other countries? Would that be a better reason to exclude it?
3. What should a paper report when one observation drives a result?

ANSWERS Should we drop Greece? Liechtenstein?

- Dropping an observation because it changes the result is choosing the answer. Greece is a real country with real data.
- A substantive reason, stated in advance, is different: in a very small state a few judges give a large number per 100,000, and the judiciary may work differently. That argues for a separate check, not for silent exclusion.
- Report the main estimate on all observations, show the estimates without the influential ones, and name them. Readers can then judge.
- Here the honest summary is: no clear relation, and the point estimate depends on one country.

PREDICT Adding a control can change the sample

- Lecture 5: theft on police and ln(GDP p.c.): police -2.04, 34 countries.
- Now add judges per 100,000 as a control. Judges are missing for Germany and Switzerland.

Predict:

1. How many countries will Stata use?
2. If the police coefficient changes, how can we tell whether the new control or the new sample is responsible?

STATA Guided Stata 3: same model, same sample

```
regress theft_100k police_100k ln_gdppc
estimates store base
regress theft_100k police_100k ln_gdppc judges_100k
estimates store judges
regress theft_100k police_100k ln_gdppc if e(sample)
estimates store base_same
estimates table base base_same judges, b(%9.2f) se(%9.2f) stats(N r2)
```

Recorded thefts per 100,000	base	base, same sample	with judges
Police per 100,000	−2.04 (1.03)	−2.08 (1.07)	−1.67 (1.04)
ln GDP per capita	439.1 (153.3)	450.2 (160.6)	472.4 (152.8)
Judges per 100,000			−14.69 (7.16)
N	34	32	32

- `e(sample)` marks the countries used by the last regression. Sample effect: $-2.04 \to -2.08$; control effect: $-2.08 \to -1.67$.

Changing samples: rules

- Every variable with missing values silently removes rows. Two columns with different N compare different countries.
- To isolate the effect of a control, estimate both models on the **same** sample (`if e(sample)`).
- Report N in every column and say which units leave the sample, and why.
- If the sample changes, ask whether the units that leave are special (small states, countries in conflict, missing statistics): the result may describe a different population.

How large is -2.04?

- Statistical significance says whether a coefficient is distinguishable from zero; it says nothing about whether it is large.
- Three ways to read the magnitude (34 countries):
  - **units**: 100 more police officers per 100,000 go with 204 fewer recorded thefts per 100,000, among countries with the same income;
  - **standard deviations**: one SD of police (123) goes with 250 fewer thefts, 0.31 SD of theft (804);
  - **elasticity at the means**: $-2.04 \times 359.5/982.9 = -0.75$: 1% more police, 0.75% fewer recorded thefts.
- Each is a description of an association, not of what hiring police would do.

DISCUSS Voigt and Wulf, all columns of Table 2

Dependent variable: de facto independence	(1)	(2)	(3)	(4)	(5)
Legal origin: English	0.24*	0.28**	0.45***	0.51***	0.54***
	(0.09)	(0.10)	(0.12)	(0.12)	(0.15)
Immunity of parliamentarians			0.23*	0.22**	0.27*
			(0.08)	(0.08)	(0.10)
Federalism: hybrid unions		−0.15		−0.14	0.02**
		(0.10)		(0.11)	(0.10)
N	74	73	61	61	51

Questions:

1. Why does *N* fall from 74 to 51?
2. The English coefficient rises from 0.28 in column (2) to 0.45 in column (3). Is it the new control or the new sample? How could the authors show it?
3. Find one number in this table that cannot be right as printed.

Voigt, S. and Wulf, A.J. (2019), *Journal of Institutional Economics* 15(1): 99–120, Table 2 (selected rows), p. 114.

ANSWERS Voigt and Wulf, all columns of Table 2

- New variables with missing values remove countries. Table 1 of the paper shows that immunity of parliamentarians is observed for 61 countries and the size of the procuracy for 51; the text does not explain the drop from 74 to 61.

- From (2) to (3) the controls change *and* N falls by 12: the table cannot tell. Re-estimating columns (1) and (2) on the 61 countries of column (3) would.

- Hybrid unions in column (5): 0.02 with standard error 0.10 gives $t = 0.2$, yet it is marked significant at 1%. Probably a typo, but a referee should ask.

- Also: footnote 10 refers to "models 2–8", but Table 2 has five models.

Writing a referee comment

1. **Where**: the exact table, column, page or footnote.
2. **What**: the issue, stated as a fact you can check ("N falls from 73 to 61 between columns 2 and 3").
3. **Why it matters**: which conclusion of the paper depends on it.
4. **What would resolve it**: a feasible check, more information, or a narrower claim.

- One comment, one issue. Main comments first, then minor ones.
- Write about the paper, not the authors: "the table does not show", not "the authors failed to".

DISCUSS Write one comment

Write one main comment on Voigt and Wulf (2019), Section 8 and Table 2, following the four steps.

- Any issue is allowed: the change of sample, the standard errors, the interpretation of a coefficient, the measures.
- Length: about 100 words.
- Then read your comment aloud and the class says which of the four steps is missing.

ANSWERS Write one comment: two examples

- **Changing samples.** "Table 2: *N* falls from 73 in column (2) to 61 in column (3), while immunity of parliamentarians is added. The English legal-origin coefficient rises from 0.28 to 0.45, and Section 8 reports it as significant in all five models. The table cannot separate the effect of the new control from that of the new sample. Please re-estimate columns (1) and (2) on the 61 countries of column (3) and list the excluded countries."

- **Standard errors.** "Footnote 10 reads a non-significant Breusch–Pagan test as showing that the model 'does not suffer from heteroscedasticity'. With 74 countries the test has limited power. Since all inference in Section 8 rests on these standard errors, please report heteroskedasticity-robust standard errors in all columns of Table 2."

Laboratory

- Worksheet: `do/exercises/06_assumptions_and_samples.do`.
- Data: `justice_europe_2019.dta` (Tasks 1–4) and `contracts_2019.dta` (Task 5).
- Task 6: rewrite the comment you wrote in class, and write one comment of the same kind on the paper you are reviewing.

Exit check

1. Robust standard errors change which numbers of a regression output, and which do they leave unchanged?
2. What is the difference between an outlier and an influential observation?
3. Two columns of a table have $N = 73$ and $N = 61$. What would you ask the authors?

Next week

- Lecture 7 (Thursday): experiments. Back to “A Fine Is a Price”, this time with the data of Table 1.
- Before lecture 7: draft your two or three comments on the paper you are reviewing, with the four steps of today.